

The Journey of a Click

How a Web Page Really Reaches Your Screen — In-Class Discovery Worksheet

Name: _____ Section: _____ Date: _____

How to use this sheet: Keep your **Terminal** and **browser** open beside you. For every task, **attempt and write your answer *before* we discuss it** — your first guesses do not need to be right. Thinking hard and being wrong is exactly how this is meant to work.

Block A — The Big Question

Q1. Write down **everything you think happens** between **A** and **B**. Capture your current thoughts, no matter how wrong — we will come back to this at the very end.

A. You open a Chrome tab, type `hotstar.com` in the search bar, and hit **Enter**.

B. You see the web page.

Block B — Finding the Server (DNS)

TRY THIS · TERMINAL

Open your Terminal and run: `ping hotstar.com` — then print the **first 2 lines** of output below. Suppose the output looks like this:

```
PING hotstar.com (99.86.30.53): 56 data bytes
64 bytes from 99.86.30.53: icmp_seq=0 ttl=250 time=224.354 ms
```

Q2. Using the output above, answer:

(a) What does `hotstar.com (99.86.30.53)` represent? _____

(b) What do the `56 data bytes / 64 bytes` represent? _____

(c) What does `time=224.354 ms` represent? _____

Q3. Why do humans prefer typing website **names** instead of **IP addresses**? And a twist: if browsers and websites were designed for **AI agents** instead of humans, would website names really be needed?

Q4. What is a **Server**? What is a **Client**? Are these **absolute** or **relative** terms?

Q5. What is a **DNS Server**? Write its full form.

Q6. Tricky one!! If the only way to find an IP were to ask the DNS server... then how would you find the **DNS server's own IP**?

Q7. What is an **ISP**?

Q8. Run `scutil --dns | grep nameserver` and print the **first 3 lines** of output here:

Q9. Why are there **multiple** outputs when you ran that command?

Q10. Some outputs have a format like `2401:4900:50:9::2e0`, and some like `192.168.1.1`. Why? What are these two formats **called**? Is there a **range of values** associated with each? (Fill the table.)

Example	What is this format called?	Typical range / notes
192.168.1.1	_____	_____
2401:4900:50:9::2e0	_____	_____

Q11. Is `192.168.1.1` a **special** address? If yes, what is special about it?

Q12. Is `8.8.8.8` another **special** address? What is it?

TRY THIS · TERMINAL

Run `ping google.com`, then run `ping 8.8.8.8`.

Q13. What is **one major difference** you notice between the two?

TRY THIS · CHROME

Open **Chrome** and visit `https://zenithschool.ai/`. In a different tab go to `chrome://net-internals/#dns` and search for `zenithschool.ai`.

Q14. What do you understand by this? Why would browser try to remember this?

Q15. So now that you have met the **browser cache**, the **OS cache**, and the **name servers** — list the full sequence of lookups that happens during DNS resolution.

1. _____ -> 2. _____ -> 3. _____ -> 4. _____

Block C — From Server to Screen

Once DNS is done and we have the IP, the browser opens a connection to that IP and sends a request — *"GET me the homepage"* — and the server sends back a **response**.

Q16. What **is** that response? Is it a **picture** of the web page, a **text file**, or something else?

Q17. What **kinds** of text files are these that the browser receives from the web server?

Q18. A million people open `amazon.com` in the same second. If there is **one machine** acting as the web server, it will crash. **How do we handle this?**

Q19. How are **HTML**, **CSS**, and **JavaScript** different from each other? (Fill the table.)

	HTML	CSS	JavaScript
Role			
Analogy			

Q20. How does the browser take all these text files and turn them into a **beautifully painted website**? So, what **is** a browser? Which is your **favourite browser** and why? Which **company** maintains your browser?

Block D — The Dynamic Web (Backend)

Q21. The home page of Amazon looks **static** — the same for everyone. But what about the orders page, or the cart page? Their content is **dynamic** — different per user, per request. There has to be a separate **program** that runs to produce that answer. **What is that?**

Q22. Which part of your **frontend** code — HTML, CSS, or JS — asks the backend for dynamic data?

Q23. What is the **connective tissue** between frontend and backend — the thing **through which** your frontend code talks to the backend?

Q24. In what **format** is the data returned from the backend to the frontend?

Q25. Write a piece of **JSON** data containing your **name**, **date of birth**, **age**, and **address**.

```
{
}
```

Q26. When you search on Amazon, **more products appear only after changing page**. Why? Why don't they show **all products at once**?

Block E — Where the Data Lives (Database)

Q27. Where should a user's data — **login credentials**, **order history**, **cart details** — be saved: in **RAM** or on the **HDD/disk**? Why?

Q28. Is it possible to store all this data **inside the application server itself**? Why / why not?

Q29. Then **where** do we store all this data?

Q30. What are the **primary responsibilities** of a database?

Block F — The Full Journey

Q31. At the very start of class, I asked you to write what happens when you type a URL. **Let's redo it** — now with everything you have learned. Trace each journey end to end.

(A) What happens from the moment you search `amazon.com` until you see the home page?

(B) What happens from the moment you search `iPhone-13` on Amazon until the moment you see the results?